Title Paper	Author	Citations	Jounral / Conference Ranking	Date	Type	Reviews
Pedestrian-Vehicle Information Modulation for Pedestrian Crossing Intention Prediction	Li Xu, Shaodi You, Gang He, Yunsong Li	20-30	IEEE Transactions on Intelligent Vehicles (Q1)	2025	Research Paper	Proposed PVIM model for pedestrian crossing intention prediction using pedestrian-vehicle interaction, spatial context, and temporal bilinear attention. Achieved 91% accuracy on PIE benchmark.
Autonomous Driving Planning Based on Large Language Model: Collaborative Driving	James Vilho, Tianhao Liang, Cong Guo, Tingting Zhang	100-150	IEEE Vehicular Technology Conference (VTC)	2025	Conference Paper	Proposed LLMDriver, a collaborative multi-agent autonomous driving framework using Large Language Models (LLMs). The system integrates memory modules, collaborative trajectory planning, Chain-of-Thought reasoning, and collision avoidance for safer autonomous driving. Evaluated on nuScenes and CARLA benchmarks, achieving strong trajectory planning performance with improved interpretability and safety in complex driving scenarios.
Action and Intention Recognition of Pedestrians in Urban Traffic	Dimitrios Varytimidis, Fernando Alonso-Fernandez, Boris Duran, Cristofer Englund	0-10	IEEE International Conference on Signal-Image Technology & Internet-Based Systems (SITIS)	2022	Conference Paper	Proposed a pedestrian action and intention recognition framework using JAAD dataset for autonomous driving environments. The study used HOG, LBP, and CNN feature extraction methods combined with SVM, ANN, k-NN, and Decision Trees to estimate pedestrian head orientation and motion.
Driving with LLMs: Fusing Object-Level Vector Modality for Explainable Autonomous Driving	Long Chen, Oleg Sinavski, Jan Hünermann, Alice Karnsund, Andrew James Willmott, Danny Birch, Daniel Maund, Jamie Shotton	0-5	IEEE International Conference on Robotics and Automation (ICRA)	2024	Conference Paper	Proposed an explainable autonomous driving framework that integrates object-level vector representations with Large Language Models (LLMs). The system combines vectorized scene information, driving context, and natural language reasoning using a multimodal LLM architecture.
MMCF-Net: A Micro-Motion Context Fusion Framework for Pedestrian Crossing Intention Recognition in Urban Scenes	Menghao Li, Weiwei Zhang, Wenfeng Guo, Hao Chen, Wangpengfei Yu, Jiejie Xu, Jun Li	35-45	IEEE Transactions on Consumer Electronics (Q1)	2025	Journal Paper	Proposed MMCF-Net, a micro-motion and context fusion framework for early pedestrian crossing intention recognition in urban environments. The model combines a High-Resolution Spatiotemporal Attention Module (HRSTA), Multi-modal Context Encoder (MCE), and Dynamic Weight Allocation (DWA) mechanism to capture subtle pedestrian behaviors such as head turning, center-of-gravity shifts, and foot micro-movements. The framework achieved 92.3% F1 score with improved robustness under occlusion, low-light, and adverse weather conditions, outperforming existing methods like PIE and PCPA.
KoMA: Knowledge-Driven Multi-Agent Framework for Autonomous Driving With Large Language Models	Kemou Jiang, Xuan Cai, Zhiyong Cui, Aoyong Li, Yilong Ren, Haiyang Yu, Hao Yang, Daocheng Fu, Licheng Wen, Pinlong Cai	50-150	IEEE Transactions on Intelligent Vehicles (Q1)	2025	Research Paper	Proposed KoMA, a knowledge-driven multi-agent autonomous driving framework powered by Large Language Models (LLMs). The framework integrates five key modules: Multi-Agent Interaction, Multi-Step Planning, Shared Memory, Ranking-Based Reflection, and Environment Modeling. KoMA enables agents to reason about surrounding vehicle intentions, perform goal-plan-action (GPA) multi-step reasoning, and share driving experiences through a shared vector memory database.
Pedestrian Trajectory Intention Prediction in Autonomous Driving Scenarios Based on Spatio-temporal Attention Mechanism	Yong Wang et al.	100-150	IEEE EIECC	2024	Research Paper	Proposed a spatio-temporal attention framework for pedestrian trajectory and intention prediction using multi-head attention and intention recognition modules, achieving 93% intention accuracy with improved real-time performance.
Pedestrian Crossing Intention Prediction via Progressive Multimodal Token Fusion for Autonomous Driving	Xiaobo Chen et al.	0-5	IEEE Transactions on Intelligent Transportation Systems (Q1)	2025	Research Paper	Proposed PedMTF, a multimodal fusion framework using pedestrian pose, bounding boxes, and vehicle speed with attention-based token fusion for accurate pedestrian crossing intention prediction

Autonomous Driving at Unsignalized Intersections: A Review of Decision-Making Challenges and Reinforcement Learning-Based Solutions	Mohammad Al-Sharman, Luc Edes, Bert Sun, Vishal Jayakumar, Hasan Tahir, Mohamed A. Daoud, Bara J. Emran, Derek Rayside, William Melek	14	IEEE Transactions on Automation Science and Engineering	2026	Journal Paper	It explains why unsignalized intersections are difficult for autonomous vehicles because of uncertainty, multi-agent interaction, partial observability, intention inference, and safety-critical decision-making. It also reviews RL and deep learning-based solutions for intersection decision-making.
Multi-Agent Deep Reinforcement Learning to Manage Connected Autonomous Vehicles at Tomorrow's Intersections	Antonio Guillen-Perez, Maria-Dolores Cano	176	IEEE Transactions on Vehicular Technology	2022	Journal Paper	It focuses on connected autonomous vehicles at intersections and proposes an autonomous intersection management system using multi-agent deep reinforcement learning. It also discusses conflict modules, priority modules, speed control, waiting time reduction, and traffic flow improvement.
Cyber Traffic Light: Safe Cooperation for Autonomous Vehicles at Dynamic Intersections	Shunsuke Aoki, Ragunathan Rajkumar	16	IEEE Transactions on Intelligent Transportation Systems	2022	Journal Paper	The paper proposes a cooperative dynamic intersection protocol using perception and V2V/V2X communication. It discusses peer-to-peer negotiation, road safety, fairness for waiting vehicles, and conflict handling at shared road segments.
Decentralized Opinion-Integrated Decision Making at Unsignalized Intersections via Signed Networks	Bhaskar Varma, Ying Shuai Quan, Karl D. von Ellenrieder, Paolo Falcone	0	arXiv / Research preprint	2026	Preprint	It proposes decentralized decision-making at unsignalized intersections without a central coordinator. Vehicles exchange intent and make GO/YIELD commitments using local interaction, which is close to V2V-style negotiation.
Autonomous Cooperation at Unsignalized Intersections Based on Multi-agent Actor-Critic Algorithm and Mixed Nash Strategy	Yalu Zhu, Shikang Lian, Panshuo Li, Wei Meng	0	CAA International Conference on Vehicular Control and Intelligence	2025	Conference Paper	It directly addresses autonomous vehicle coordination at unsignalized intersections. It combines multi-agent learning with game-theoretic acceleration/deceleration intentions and shows collision-free driving in SUMO.
GAMEOPT: Optimal Real-time Multi-Agent Planning and Control at Dynamic Intersections	Nilesh Suriyarachchi, Rohan Chandra, John S. Baras, Dinesh Manocha	25	arXiv / Research preprint	2022	Preprint	It proposes multi-agent planning for unsignalized intersections and compares throughput, travel time, fuel consumption, and fairness. This supports the need to measure waiting time and fairness in your component.
GamePlan: Game-Theoretic Multi-Agent Planning with Human Drivers at Intersections, Roundabouts, and Merging	Rohan Chandra, Dinesh Manocha	83	arXiv / Research preprint	2022	Preprint	It focuses on multi-agent planning at unsignaled intersections, roundabouts, and merging. It uses driving style and game theory to decide turn order and avoid collisions and deadlocks. Useful for explaining right-of-way negotiation and driver intent.
Risk-Aware Reinforcement Learning for Cooperative Autonomous Vehicle Coordination with Adaptive Risk Sensitivity and Multi-Agent Optimization	Malikireddy Ramesh Reddy, Annalakshmi Govindaraj	0	EAI Endorsed Transactions on Internet of Things	2026	Journal Paper	Supporting literature for safety and risk-aware cooperative AV coordination. It evaluates autonomous vehicle coordination using SUMO, CARLA, NGSIM, and INTERACTION, and reports improvements in collision reduction, throughput, robustness, and cooperative behaviour.
A Vehicle-Infrastructure Multi-layer Cooperative Decision-making Framework	Yiming Cui, Shiyu Fang, Peng Hang, Jian Sun	2	IEEE Intelligent Vehicles Symposium	2025	Conference Paper	Related, but less direct than the V2V-only papers because it includes V2I and infrastructure support. It is useful for comparing your decentralized/V2V-style approach with infrastructure-assisted cooperative decision-making at unsignalized intersections.
PIE: A Large-Scale Dataset and Models for Pedestrian Intention Estimation and Trajectory Prediction	Amir Rasouli, Iuliia Kotseruba, Toni Kunic, John K. Tsotsos	637	ICCV / Top-tier Computer Vision Conference	2019	Dataset + Intention / Trajectory Prediction	Introduces the PIE dataset for pedestrian intention estimation and trajectory prediction. Shows that AV systems must understand pedestrian intentions before actions are already in progress; reports 79% intention accuracy and improved trajectory prediction when intention is combined with motion.
MetaMind: Modeling Human Social Thoughts with Metacognitive Multi-Agent Systems	Xuanming Zhang, Yuxuan Chen, Samuel Yeh, Sharon Li	22	NeurIPS 2025 / arXiv	2025	Theory of Mind + Multi-Agent Reasoning	Proposes a staged ToM-inspired multi-agent framework: a Theory-of-Mind Agent infers hidden mental states, a Moral Agent refines them using cultural/ethical constraints, and a Response Agent validates the final response. Reports 35.7% improvement in real social scenarios and gains in ToM reasoning.
PSI: A Pedestrian Behavior Dataset for Socially Intelligent Autonomous Car	Renran Tian, Yaobin Chen, Joshua Domeyer, Heishiro Toyoda, Rini Sherony, Zhengming Ding	53	arXiv / AV Dataset Paper	2021	Dataset + Explainable Social Intelligence	Introduces PSI dataset with dynamic pedestrian intent-change labels and text explanations of driver reasoning. Emphasizes that future AVs need both technical and social capabilities for pedestrian interaction.
Pedestrian Behavior Prediction Using Deep Learning Methods for Urban Scenarios: A Review	Chi Zhang, Christian Berger	106	IEEE Transactions on Intelligent Transportation Systems / Q1	2023	Review	Comprehensive review of pedestrian behavior prediction for automated driving, covering trajectory prediction, intention prediction, datasets, inputs, features, metrics, and model structures such as LSTM, CNN, GCN, and Transformers.

Multi-scale pedestrian intent prediction using 3D joint information as spatio-temporal representation	Sarfraz Ahmed, Ammar Al Bazi, Chitta Saha, Sujan Rajbhandari, M. Nazmul Huda	31	sciencedirect-Expert Systems With Applications/ Q1	2023	Pose Estimation + LSTM	Uses pose-estimated keypoints accumulated over video frames as spatio-temporal input to an LSTM for classifying pedestrian crossing behavior. Evaluates on JAAD and PIE and reports up to 94% accuracy with 6.1 ms runtime.
Predicting Road Crossing Behaviour using Pose Detection and Sequence Modelling	Subhasis Dasgupta, Jaydip Sen, Preetam Saha, Agniva Roy	0	IEEE	2025	MediaPipe + LSTM / GRU / 1D CNN	models to predict road-crossing intent. Compares LSTM, GRU, and 1D CNN; reports LSTM 87.00% accuracy, GRU 85.52% accuracy with higher AUC, and faster inference for 1D CNN.
MediaPipe: A Framework for Building Perception Pipelines	Camillo Lugaresi et al.	2748	Google Research / arXiv	2019	Pose / Perception Framework	Introduces MediaPipe as a graph-based framework for building perception pipelines over sensory data such as video. Designed for reusable components, rapid prototyping, performance evaluation, and deployment across platforms.
PedAST-GCN: Fast Pedestrian Crossing Intention Prediction Using Spatial–Temporal Attention Graph Convolution Networks	Yancheng Ling, Zhenliang Ma, Qi Zhang, Bangquan Xie, Xiaoxiong Weng	36	IEEE Transactions on Intelligent Transportation Systems / Q1	2024	GCN + Attention + Multimodal Features	Proposes a lightweight GCN with spatial-temporal and modality attention using pedestrian pose, bounding box, and ego-vehicle speed. Validated on JAAD and PIE with strong accuracy and computation-time performance.
GTransPDM: A Graph-embedded Transformer with Positional Decoupling for Pedestrian Crossing Intention Prediction	Chen Xie, Ciyun Lin, Xiaoyu Zheng, Bowen Gong, Antonio M. López	12	arXiv	2025	Graph Transformer + Multimodal Fusion	Uses a positional decoupling module, graph-embedded Transformer, pose skeletons, pedestrian position, and ego-vehicle motion for pedestrian crossing intention prediction. Reports 92% accuracy on PIE and 87% on JAAD with 0.05 ms processing speed.
Diving Deeper Into Pedestrian Behavior Understanding: Intention Estimation, Action Prediction, and Event Risk Assessment	Amir Rasouli, Iuliia Kotseruba	4	arXiv / Huawei + York University	2024	Benchmark + Risk Assessment	Clarifies the difference between intention estimation, action prediction, and event risk assessment using JAAD and PIE. Shows these tasks are complementary and should not be treated as the same safety problem.
Pedestrian Intention Prediction via Vision-Language Foundation Models	Authors not clearly visible in uploaded snippet	5	arXiv / Vision-Language Model Paper	2025	Vision-Language Foundation Models	Uses vision-language foundation models and hierarchical prompt templates for pedestrian crossing intention prediction. Shows that adding behavioral cues and ego-vehicle speed/dynamic context improves performance.
Pedestrian Intention and Trajectory Prediction in Unstructured Traffic Using IDD-PeD	Ruthvik Bokkasam, Shankar Gangisetty, A. H. Abdul Hafez, C. V. Jawahar	2	arXiv	2025	Dataset + Intention / Trajectory Prediction	Introduces IDD-PeD for unstructured traffic with occlusion, illumination changes, unsignalized scenes, and vehicle-pedestrian interactions. Shows strong performance drops for existing PIP/PTP baselines in chaotic traffic.
REACT: Runtime-Enabled Active Collision-avoidance Technique for Autonomous Driving	Heye Huang, Hao Cheng, Zhiyuan Zhou, Zijin Wang, Qichao Liu, Xiaopeng Li	11	Autonomous Driving / Risk Assessment Paper	2025	Runtime Risk + Collision Avoidance	Proposes a runtime risk assessment and active avoidance framework using interaction-aware risk fields, hierarchical warning levels, and interpretable avoidance outputs. Compares TTC, THW, RSS, DRF, and REACT.
Interaction-Aware Model Predictive Decision-Making for Socially-Compliant Autonomous Driving in Mixed Urban Traffic Scenarios	Balint Varga, Thomas Brand, Marcus Schmitz, Ehsan Hashemi	6	arXiv / Socially-Compliant Autonomous Driving	2025	MPC + Pedestrian Intention	Combines a gap-acceptance-inspired pedestrian intention model with MPC. Maps TTC to continuous crossing likelihood and models hesitation/standstill behavior for socially compliant AV decision-making.
Predict Vehicle Collision by TTC from Motion Using Single Video Camera	Mehmet Kilicarslan, Jiang Yu Zheng	99	IEEE Transactions on Intelligent Transportation Systems	2019	TTC / Collision Risk	Computes Time-to-Collision from motion divergence in monocular driving video without object recognition or depth sensing. Shows collision danger can be estimated from visual size/motion changes.
Safety improvement of Autonomous emergency braking (AEB) using STPA & FBD methods	Islem Berri, Youcef Zennir, Khalid Larit	0	International Journal of Automation and Safety	2025	AEB + Safety Analysis	Combines STPA and FBD methods for AEB safety analysis. Describes AEB architecture, perception, decision/control, actuation, and TTC-based intervention stages.
Pedestrian Models for Autonomous Driving Part II: High-Level Models of Human Behavior	Fanta Camara et al.	208	IEEE Transactions on Intelligent Transportation Systems / Survey	2020	Survey + High-Level Behavior Models	Reviews high-level pedestrian behavior models for AVs, including trajectory prediction, interaction, psychological/social models, game theory, and signaling. States that AVs require psychological and social understanding at higher autonomy levels.
Predicting Pedestrian Violations in Urban Intersections: A Comparison of Random Forest and XGBoost Models	Stella Roussou, Apostolos Ziakopoulos, George Yannis	0	Safety Modelling / Conference Paper	2025	Pedestrian Violation Prediction	Compares Random Forest and XGBoost for illegal pedestrian crossing prediction using trajectory paths, vehicle data, traffic light status, proximity, and TTC features. Shows model suitability depends on environmental complexity.
Argos: A Decentralized Federated System for Detection of Traffic Signs in CAVs	Seyed Mahdi Haji Seyed Hossein, Alireza Hosseini, Amirali Shahriary, Soheil Hajian Manesh	0	Peer-reviewed Conference Paper	2025	Conference	Proposed decentralized federated learning framework for traffic sign detection in CAVs. FedProx handled heterogeneous data better than FedAvg and FedAdam. Increasing server rounds improved accuracy from below 0.1 to above 0.8. Gap: No intelligent knowledge selection or novelty-aware sharing

Policy Search, Retrieval, and Composition via Task Similarity in Collaborative Agentic Systems (MOSAIC)	Saptarshi Nath, Christos Peridis, Eseoghene Benjamin, Xinran Liu, Soheil Kolouri,	8	Peer-reviewed Conference Research Paper	2025	Conference	ntroduced MOSAIC collaborative agent learning framework using Wasserstein task embeddings for selective policy sharing. Gap: No centralized verification layer; incorrect policies may propagate.
FLAD: Federated Learning for LLM-based Autonomous Driving in Vehicle-Edge-Cloud Networks	Tianao Xiang, Mingjian Zhi, Yuanguo Bi, Lin Cai, Yuhao Chen	1	Conference-style Research Paper / arXiv Preprint	2025	Research Paper	Proposed cloud-edge-vehicle collaborative federated learning for autonomous driving. Introduced SWIFT DQN scheduling and federated hybrid parallelism. Gap: High memory requirements and no novelty-aware sharing.
Federated Reinforcement Learning: Techniques, Applications, and Open Challenges	Jiaju Qi, Qihao Zhou, Lei Lei, Kan Zheng	338	Peer-reviewed Journal / Survey	2021	Review	Comprehensive survey on Federated Reinforcement Learning (FRL). Discussed communication efficiency, privacy, and distributed RL challenges. Gap: Lacks adaptive intelligent sharing.
Traffic Sign Recognition with Multi-Scale Convolutional Networks	Pierre Sermanet, Yann LeCun	3000	Peer-reviewed Conference (IJCNN, Q1)	2021	Journal	Achieved 99.17% accuracy on the GTSRB dataset using multi-scale CNNs. Established an important benchmark for traffic sign recognition.
Traffic Sign Recognition and Detection using YOLOv8	Shevtekar, Kulkarni	10	Peer-reviewed Journal	2024	Journal	Achieved approximately 95% mAP using YOLOv8 for traffic sign detection and multi-class recognition. Suitable for autonomous driving and ADAS applications
Enhancing Traffic Sign Recognition On The Performance Based On YOLOv8	Baba Ibrahim, Zhou Kui	0	arXiv Preprint	2025	Thesis / Research Study	Enhanced YOLOv8 using CA, BiFPN, ODConv, and LSKA modules. Improved robustness under adverse conditions and edge deployment. Supports small object detection and visual prominence for traffic signs.
Federated Learning for Connected and Automated Vehicles: Survey	Chellapandi, Yuan, Brinton, Zak, Wang	82	Peer-reviewed IEEE Journal	2024	Review	Surveyed federated learning techniques for connected and automated vehicles with focus on scalability and heterogeneity.
Beyond Context Sharing: A Unified Agent Communication Protocol (ACP) for Secure, Federated, and Autonomous Agent-to-Agent (A2A) Orchestration	Naveen Krishnan	0	arXiv Preprint	2026	Preprint	Proposed decentralized A2A communication protocol for agentic AI systems with federated orchestration and secure communication.